Existence of the set-to-set distance exponent

In the white-noise setup of Ding–Gwynne–Zhuang

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Abstract

Fix a dimension $d \geq 2$ and a parameter $\psi > 0$. For the finite-range white-noise regularization with the normalized unit-ball indicator kernel, we prove that the median distance between any two disjoint compact sets with nonempty interiors has a deterministic exponential rate. The rate is the same for all such pairs. This establishes the existence of the quantity $\widetilde{Q}(\psi)$ defined in equation (1.9) of *Percolation of thick points of the log-correlated Gaussian field in high dimensions*, arXiv:2410.22075.

The main estimate is almost-supermultiplicativity of annulus-crossing medians. A logarithmic gap between coarse and fine scales makes each exceptional fine-scale cluster small relative to the variation scale of the coarse field. An arbitrary fine path then gives a coarse polygonal path by replacing its passages through these clusters with straight segments. A finite endpoint cover and similarities transfer the resulting exponent to the unrestricted set-to-set distances in the stated definition.

1 Statement and setup

Let W be white noise on $\mathbb{R}^d \times (0, \infty)$ and put

$$K(x) = |B_1(0)|^{-1/2} \mathbf{1}_{\{|x|_2 < 1\}}.$$

For real numbers $a < b$, define

$$H_{a,b}(x) = \int_{\mathbb{R}^d} \int_{2^{-b}}^{2^{-a}} K\left(\frac{y-x}{t}\right) t^{-(d+1)/2} \, W(dy, dt). \tag{1}$$

Set $H_{a,a} = 0$. Write $h_n = H_{0,n}$ for integers $n \geq 1$ and $h_0 = 0$. These fields have continuous modifications, which we use throughout. The parameter denoted by ξ in [1] is denoted here by ψ.

For a positive continuous density w, let

$$\mathcal{D}_w(A, B; U) = \inf_P \int_0^1 w(P(t)) |P'(t)|_2 \, dt,$$

where the infimum is over piecewise continuously differentiable paths from A to B whose images lie in U. Closed convex domains are allowed in this notation. Omit U when $U = \mathbb{R}^d$. Write $D_n = \mathcal{D}_{e^{\psi h_n}}$ and, for a field f, write $D_f = \mathcal{D}_{e^{\psi f}}$. Thus $\mathcal{D}$ takes a density as its subscript, whereas D_f takes a field.

Theorem 1.1. *For every $d \geq 2$ and $\psi > 0$, there is a finite deterministic number $\widetilde{Q}(\psi) \in \mathbb{R}$ such that, for every two disjoint compact sets $K_1, K_2 \subset \mathbb{R}^d$ with nonempty interiors,*

$$\operatorname{Med} D_n(K_1, K_2) = 2^{-(1-\psi\widetilde{Q}(\psi))n+o(n)}. \tag{2}$$

The same rate holds for every fixed quantile in $(0,1)$. For each fixed pair K_1, K_2, under the white-noise coupling above,

$$\frac{\log D_n(K_1, K_2)}{n \log 2} \longrightarrow -1 + \psi\widetilde{Q}(\psi) \qquad \textit{almost surely.} \tag{3}$$

All logarithms without a subscript are natural. Constants may depend on d and ψ and on fixed domains or fixed probability levels, but not on scale indices. Different occurrences of C may denote different constants.

For $r > 0$ and $z \in \mathbb{R}^d$, let

$$C_r(z) = z + [-r, r]^d, \qquad C_r = C_r(0).$$

Our reference crossing is

$$T_n = D_n(C_1, \partial C_2; C_2), \qquad b_n = \operatorname{Med} T_n, \qquad a_n = \log b_n. \tag{4}$$

We take $b_0 = 1$ and $a_0 = 0$. A path can be stopped at its first hit of ∂C_2, and its starting portion can be removed up to its last visit to C_1. Thus the same infimum is obtained by crossings of the closed shell between ∂C_1 and ∂C_2. It also equals the unrestricted distance from C_1 to ∂C_2. These observations are used only for this particular annular distance.

The proof has two main steps:

$$a_{m+n} \geq a_m + a_n - C(m+n)^{2/3}, \tag{5}$$

$$|\log \operatorname{Med} D_n(K_1, K_2) - a_n| \leq C_{K_1,K_2}\sqrt{n}. \tag{6}$$

The first gives convergence of a_n/n; the second identifies the exact quantity in (2). Both are proved below.

2 Gaussian estimates

We record the estimates needed for the nonsmooth kernel K. In particular, no differentiability of h_n is used.

Lemma 2.1 (Field bounds)**.** *For $a < b$ one has*

$$\operatorname{Var} H_{a,b}(x) = (b-a)\log 2, \qquad \mathbb{E}\big[(H_{a,b}(x) - H_{a,b}(y))^2\big] \leq C_d 2^b |x-y|_2. \tag{7}$$

Fix $R < \infty$ and $\varepsilon \in (0,1)$. For integers $0 \leq a < b$,

$$\mathbb{P}\left[\sup_{C_R} |H_{a,b}| \leq C_{R,\varepsilon}\sqrt{(b-a)(b+1)}\right] \geq 1 - \varepsilon. \tag{8}$$

For integers $m \geq 0$ and $0 < \delta \leq 2^{-m}$,

$$\mathbb{P}\left[\sup_{\substack{x,y \in C_R \\ |x-y|_\infty \leq \delta}} |h_m(x) - h_m(y)| \leq C_{R,\varepsilon}\sqrt{2^m \delta \log(2 + \delta^{-1})}\right] \geq 1 - \varepsilon. \tag{9}$$

Proof. The normalization $\int K^2 = 1$ gives the variance formula. The symmetric difference of two unit balls whose centers differ by u has volume at most $C_d \min\{|u|_2, 1\}$. Therefore

$$\|K(\cdot - u) - K\|_{L^2}^2 \le C_d \min\{|u|_2, 1\},$$

and the white-noise isometry gives

$$\mathbb{E}[(H_{a,b}(x) - H_{a,b}(y))^2] \le C_d |x - y|_2 \int_{2^{-b}}^{2^{-a}} \frac{dt}{t^2} \le C_d 2^b |x - y|_2.$$

Gaussian moments of arbitrarily high order and Kolmogorov's criterion give a continuous modification on every compact set.

For completeness, the supremum estimates follow from local Gaussian entropy and a finite covering. On a cube of side 2^{-b}, the process $H_{a,b}(x) - H_{a,b}(x_0)$, after rescaling the cube to unit size, has canonical distance at most $C_d|u - v|_2^{1/2}$. Its covering number at canonical radius s is at most $C_d s^{-2d}$ for small s. The Gaussian entropy integral $\int_0^1 \sqrt{\log(C_d s^{-2d})}\, ds$ is finite. Hence its expected supremum in absolute value is bounded by a constant, and Gaussian concentration gives a Gaussian upper tail with a constant variance parameter. There are at most $C_R 2^{db}$ cubes in the covering of C_R. A union bound controls their largest oscillation by $C_{R,\varepsilon}\sqrt{b+1}$. The values at the cube centers have variance $(b-a)\log 2$, so another union bound controls their maximum absolute value by $C_{R,\varepsilon}\sqrt{(b-a)(b+1)}$. Since $b - a \ge 1$, this proves (8).

To prove (9), cover C_R by cubes of side δ and enlarge each by a fixed factor. Every pair at sup-norm distance at most δ lies in one of these enlarged cubes. On such a cube the increment process $h_m(x) - h_m(x_0)$ has expected absolute supremum at most $C\sqrt{2^m \delta}$ and a Gaussian tail with variance parameter $C 2^m \delta$, by the same entropy calculation. A union bound over at most $C_R \delta^{-d}$ cubes gives (9). The case $m = 0$ is immediate. □

Lemma 2.2 (Concentration of logarithmic distances)**.** *Let X_n be a distance between fixed disjoint nonempty compact sets, either unrestricted, in an open domain containing the endpoint sets, or in a compact convex domain containing the endpoint sets. In the open case assume that a finite polygonal connection exists. The instances used below include T_n. If M_n is a median of X_n, then*

$$\mathbb{P}\left[|\log X_n - \log M_n| > t\right] \le 2\exp\left(-\frac{t^2}{2\psi^2 n \log 2}\right), \qquad t > 0, n \ge 1. \tag{10}$$

In particular, changing a fixed quantile changes the logarithm of the distance by at most $C\sqrt{n}$.

Proof. A Cameron–Martin shift of white noise by $u \in L^2$ changes the field at every point by at most $\sqrt{n \log 2}\, \|u\|_{L^2}$, by Cauchy–Schwarz applied to (1). This bound is uniform on all of $\mathbb{R}^d$. Comparing the length of every admissible path shows that $\log X_n$ is $\psi\sqrt{n \log 2}$-Lipschitz in the Gaussian Cameron–Martin norm. The Gaussian isoperimetric inequality, applied to its median sublevel set and superlevel set, gives (10).

Here the logarithm is finite and integrable, including for unrestricted distances. Indeed, take a bounded neighborhood V of the first compact set of width smaller than its Euclidean distance to the second set. Every connecting path travels a fixed positive Euclidean length inside V, giving a lower bound of the form $c\exp(\psi \inf_V h_n)$. A fixed polygonal connection gives an upper bound $C\exp(\psi \sup_{V'} h_n)$ on another bounded set V'. The Gaussian suprema have finite expectations. Measurability follows by using a countable dense family of polygonal paths and dense subsets of the endpoint sets; continuity of the field gives the same infimum. The same Lipschitz argument can also be obtained by finite-dimensional approximation, as in [2, Lemma 2.7]. □

Two simple consequences will be used repeatedly. For every fixed $p \in (0,1)$ there is $c_p < \infty$ such that

$$\mathbb{P}[T_n < b_n e^{-c_p\sqrt{n}}] \le p. \tag{11}$$

Also, if an upper bound $X_n \le L_n$ has any fixed positive probability p, then

$$\operatorname{Med} X_n \le L_n e^{C_p\sqrt{n}}. \tag{12}$$

For the latter assertion, choose C_p so that the lower-tail bound in (10) is strictly less than p.

Lemma 2.3 (Bounds and scale shifts for a_n)**.** *There is $C < \infty$ such that*

$$|a_n| \le Cn \qquad (n \ge 1), \tag{13}$$

and for integers $q \ge 0$, $k \ge 1$,

$$|a_{q+k} - a_q| \le C\sqrt{k(q+k+1)}. \tag{14}$$

Proof. Every reference crossing has Euclidean length at least 1, and one such crossing is a line segment of length 1. Consequently,

$$e^{-\psi \sup_{C_2}|h_n|} \le T_n \le e^{\psi \sup_{C_2}|h_n|}.$$

Equation (8) with $(a,b) = (0,n)$ proves (13) by using a probability greater than $1/2$.

Under the same white noise,

$$e^{-\psi S} T_q \le T_{q+k} \le e^{\psi S} T_q, \qquad S = \sup_{C_2} |H_{q,q+k}|.$$

With probability at least 0.99, (8) bounds S by $C\sqrt{k(q+k+1)}$. With probability at least 0.99, (10) bounds $|\log T_q - a_q|$ by $C\sqrt{q}$; for $q = 0$ this difference is zero. On their intersection, $\log T_{q+k}$ lies within $C\sqrt{k(q+k+1)}$ of a_q. The intersection has probability at least 0.98, so every median lies in the same interval. This proves (14). □

3 A deterministic comparison for arbitrary paths

The following lemma is the geometric part of the proof. It is uniform over all paths. In particular, the path may depend arbitrarily on both densities.

Lemma 3.1 (Replacing passages through exceptional clusters)**.** *Let $\Omega \subset \mathbb{R}^d$ be compact and convex, and let w, v be positive continuous densities on a neighborhood of Ω and of the cubes considered below. Fix $r, a > 0$, an integer $L \ge 1$, and $\eta \ge 0$. Let $\mathcal{Z} \subset r\mathbb{Z}^d$ contain every nearest grid point to a point of Ω. Classify its sites as good or bad, and join two sites when their sup-norm distance is at most $3r$. Assume:*

(i) Every component of bad sites in the induced graph on $\mathcal{Z}$ has at most L vertices.

(ii) For every good $z \in \mathcal{Z}$,

$$\mathcal{D}_v(C_r(z), \partial C_{2r}(z); C_{2r}(z)) \ge ra.$$

(iii) For $z \in \mathcal{Z}$ and $u \in \Omega$ with $|u - z|_\infty \le 8(L+1)r$,

$$e^{-\eta} w(z) \le w(u) \le e^{\eta} w(z).$$

Then, whenever $x, y \in \Omega$ and $|x-y|_\infty > 6(L+1)r$,

$$\mathcal{D}_{wv}(x, y; \Omega) \geq \frac{ae^{-2\eta}}{12\sqrt{d}\,(L+1)} \mathcal{D}_w(x, y; \Omega). \tag{15}$$

Proof. Fix a piecewise continuously differentiable path $\gamma : [0, T] \to \Omega$ from x to y. Construct successive times and grid centers as follows. Set $\tau_0 = 0$, and let z_i be a nearest point of $r\mathbb{Z}^d$ to $\gamma(\tau_i)$, so that $|\gamma(\tau_i) - z_i|_\infty \leq r/2$. The next time τ_{i+1} is the first hit of $\partial C_{2r}(z_i)$ after τ_i. Stop if there is no such hit before the terminal time. Call an interval $[\tau_i, \tau_{i+1}]$ completed; there is at most one final incomplete interval.

Each completed interval has Euclidean length at least $3r/2$. Thus there are finitely many of them. Consecutive centers satisfy $|z_{i+1} - z_i|_\infty \leq 5r/2$, and hence are adjacent in the graph. For a completed interval with a good center, (ii) and (iii) give

$$\mathrm{len}_{wv}(\gamma|_{[\tau_i, \tau_{i+1}]}) \geq rae^{-\eta} w(z_i). \tag{16}$$

Indeed, that segment starts in $C_r(z_i)$, stays in $C_{2r}(z_i)$, and ends on its boundary.

Let $i_1 < \cdots < i_M$ be all completed good indices, including repeated visits to the same good site if there are any, and put $u_j = \gamma(\tau_{i_j})$. Every maximal run of completed bad centers lies in a single bad component. Its diameter is at most $3(L-1)r$, regardless of how often the path revisits its sites. It follows that consecutive points in the list

$$x, u_1, \ldots, u_M, y \tag{17}$$

are at sup-norm distance at most $6(L+1)r$. More explicitly, the initial portion is at most $(3L + 1/2)r$; a segment between good starts with an intervening bad run is at most $(3L+3)r$; and the terminal portion with a bad run is at most $(3L + 9/2)r$. The last bound includes the final incomplete interval. Without an intervening bad run, the corresponding bounds are $7r/2$ and $5r$, respectively.

There must be a completed good index. Otherwise all completed centers are bad and lie in one component, and the same diameter bound, including the initial and terminal offsets, gives $|x-y|_\infty \leq 6(L+1)r$, a contradiction.

Join the points in (17) by straight segments. Convexity keeps them in Ω. Charge each outgoing segment from u_j to z_{i_j}, and charge the initial segment from x to u_1 to z_{i_1} as well. Each charged segment has Euclidean length at most $6\sqrt{d}(L+1)r$ and stays within sup-norm distance $8(L+1)r$ of its charged center. By (iii), the resulting polygonal path has w-length at most

$$12\sqrt{d}(L+1)re^{\eta} \sum_{j=1}^{M} w(z_{i_j}).$$

On the other hand, the completed intervals in (16) are disjoint in time, so

$$\mathrm{len}_{wv}(\gamma) \geq rae^{-\eta} \sum_{j=1}^{M} w(z_{i_j}).$$

Their spatial traces need not be disjoint. Combining these two inequalities and taking the infimum over γ gives (15). □

4 Almost-supermultiplicativity of crossing medians

Proposition 4.1. *There is $C < \infty$ such that for all integers $m, n \geq 1$,*

$$a_{m+n} \geq a_m + a_n - C\sqrt{(m+n)\log(m+n+2)}. \tag{18}$$

In particular, (5) *holds after increasing C.*

Proof. Write $N = m + n$. By symmetry in the desired inequality, assume $m \leq n$. Choose a fixed integer k_0, sufficiently large depending only on d, and set

$$k = k_0 + 8\lceil \log_2(N+2) \rceil, \qquad j = m + k, \qquad \ell = n - k, \qquad r = 2^{-j}. \tag{19}$$

For all sufficiently large N, $n \geq N/2 > k$ and thus $\ell \geq 1$. The finitely many remaining values of N will be absorbed at the end. Decompose

$$h_N = H + G + F, \qquad H = h_m, \quad G = H_{m,j}, \quad F = H_{j,N}. \tag{20}$$

Fine-scale exceptional clusters. Let $\mathcal{Z} = r\mathbb{Z}^d \cap C_3$, with graph adjacency at sup-norm distance at most $3r$. It contains all nearest grid points to C_2. For a number $c_* > 0$ to be fixed, call $z \in \mathcal{Z}$ good if

$$D_F(C_r(z), \partial C_{2r}(z); C_{2r}(z)) \geq r b_\ell e^{-c_*\sqrt{\ell}}, \tag{21}$$

and bad otherwise. Here D_F uses density $e^{\psi F}$. White-noise scaling gives the exact identity in law

$$r^{-1} D_F(C_r(z), \partial C_{2r}(z); C_{2r}(z)) \overset{\text{law}}{=} T_\ell. \tag{22}$$

By (11), c_* can be chosen so that the bad-site probability is at most any prescribed constant $p_* > 0$.

The event at z uses the fine field only on $C_{2r}(z)$. Since the time parameter of F is at most r and K is supported in the Euclidean unit ball, it is measurable with respect to white noise with spatial coordinate in $C_{3r}(z)$. Therefore all bad-site events in one residue class of $r\mathbb{Z}^d$ modulo 7 are mutually independent. There are 7^d such classes.

The graph degree is at most $\Delta = 7^d - 1$. The number of connected vertex sets of size s containing a specified root is at most Δ^{2s}: a depth-first traversal of a spanning tree has at most $2s$ steps. Every such set contains at least $s/7^d$ vertices in one residue class. Thus the probability that all its vertices are bad is at most $p_*^{s/7^d}$. Choose p_* so small that

$$\Delta^2 p_*^{1/7^d} \leq e^{-2}.$$

There are at most $C_d 2^{dj}$ sites in $\mathcal{Z}$. A component of size at least s contains a connected set of size exactly s. The union bound therefore yields

$$\mathbb{P}[\text{some bad component has at least } s \text{ vertices}] \leq C_d 2^{dj} e^{-2s}. \tag{23}$$

Consequently one can choose A_d so that, with

$$L = \lceil A_d(j+1) \rceil, \tag{24}$$

all bad components have at most L vertices with probability at least 0.99. Denote this event by E_1.

Coarse and intermediate fields. Set $\delta = 8(L+1)r$. By (19)–(24),

$$2^m\delta = 8(L+1)2^{-k}, \qquad L \le C_d(N+1)$$

for large N. Moreover, $\log(2+\delta^{-1}) \le C_d(N+1)$. Thus

$$2^m\delta\log(2+\delta^{-1}) \le C_d(N+1)^2 2^{-k}.$$

For all sufficiently large N, $\delta \le 2^{-m}$, $6(L+1)r < 1$, and (9) with $R = 4$ gives an event E_2 of probability at least 0.99 on which

$$|H(x) - H(y)| \le 1 \quad \text{for } x, y \in C_4,\ |x-y|_\infty \le \delta. \tag{25}$$

Increasing k_0 is also possible; only fixed constants are involved. By (8), there is an event E_3 of probability at least 0.99 on which

$$\sup_{C_2} |G| \le C\sqrt{k(j+1)}. \tag{26}$$

Finally, (10) gives an event E_4 of probability at least 0.99 on which

$$T_m \ge b_m e^{-C\sqrt{m}}. \tag{27}$$

No independence between these four events is needed. Their intersection has probability at least 0.96.

Comparison of every crossing path. On $E_1 \cap E_2$, apply Lemma 3.1 with

$$\Omega = C_2, \qquad w = e^{\psi H}, \qquad v = e^{\psi F}, \qquad a = b_\ell e^{-c_*\sqrt{\ell}}, \qquad \eta = \psi.$$

All pairs $x \in C_1$, $y \in \partial C_2$ have sup-norm separation at least 1, so the lemma applies. Taking the infimum over these endpoints gives

$$\mathcal{D}_{e^{\psi(H+F)}}(C_1, \partial C_2; C_2) \ge \frac{b_\ell e^{-c_*\sqrt{\ell}-2\psi}}{12\sqrt{d}(L+1)} T_m.$$

On E_3, the additional density $e^{\psi G}$ is everywhere at least $\exp[-C\sqrt{k(j+1)}]$ on C_2. Combining this with E_4, we get on an event of probability at least 0.96,

$$T_N \ge b_m b_\ell \exp\left[-C\left(\sqrt{m} + \sqrt{\ell} + \sqrt{k(j+1)} + \log(L+1) + 1\right)\right]. \tag{28}$$

In particular, every median b_N satisfies the same deterministic lower bound. Since $k = O(\log(N+2))$, $j \le N$ and $L = O(N+1)$,

$$a_N \ge a_m + a_\ell - C\sqrt{N\log(N+2)}. \tag{29}$$

By Lemma 2.3 and $n = \ell + k$,

$$a_\ell \ge a_n - C\sqrt{k(n+1)} \ge a_n - C\sqrt{N\log(N+2)}.$$

This proves (18) for all sufficiently large N. For the finitely many remaining splits, enlarge C, using the finiteness of every a_q. Finally, $\sqrt{N\log(N+2)} \le CN^{2/3}$ for $N \ge 1$. □

5 Convergence of the annulus exponent

We include the elementary almost-superadditive argument so that no subadditivity theorem with implicit hypotheses is required.

Lemma 5.1 (Almost-superadditivity)**.** *Suppose $u_0 = 0$, $|u_n| \le C_0 n$, and for integers $m, n \ge 1$,*

$$u_{m+n} \ge u_m + u_n - C_1(m+n)^\alpha$$

for some $0 < \alpha < 1$. Then u_n/n converges to a finite limit.

Proof. Fix $r \ge 1$. Repeatedly combine two equal blocks. For $j \ge 0$,

$$\begin{aligned} u_{2^j r} &\ge 2^j u_r - C_1 2^j r^\alpha \sum_{i=1}^{j} 2^{-(1-\alpha)i} \\ &\ge 2^j u_r - C_2 2^j r^\alpha, \end{aligned}$$

where C_2 is independent of r and j. For an integer $q \ge 1$, write q in binary, apply this estimate to its nonzero dyadic blocks, and merge those blocks. There are at most $1 + \log_2 q$ blocks, and every merged total is at most qr. Thus

$$u_{qr} \ge q u_r - C_2 q r^\alpha - C_3 (qr)^\alpha \log(q+2).$$

Write $N = qr + s$ with $0 \le s < r$. If $s > 0$, apply the assumed inequality once more to add u_s; if $s = 0$, do nothing. For this fixed r, divide by N and let $N \to \infty$. Since $\alpha < 1$, the merging errors divided by N vanish, and $u_s/N \to 0$ uniformly over the finitely many possible remainders. Therefore

$$\liminf_{N\to\infty} \frac{u_N}{N} \ge \frac{u_r}{r} - C_2 r^{\alpha-1}.$$

The linear bound makes the limsup finite. Choose $r \to \infty$ along a sequence realizing that limsup. The error tends to zero, so the liminf is at least the limsup. □

By (13), Proposition 4.1, and Lemma 5.1 with $\alpha = 2/3$, there is a finite deterministic $\lambda = \lambda(d, \psi)$ such that

$$\lim_{n\to\infty} \frac{a_n}{n} = \lambda, \qquad b_n = e^{\lambda n + o(n)}. \tag{30}$$

6 Comparison with arbitrary compact sets

This section treats the unrestricted distances in (2). Only distances in bounded domains are compared by scaling a field; concentration is then applied to the unrestricted target distance.

Lemma 6.1 (A fixed similarity changes logarithmic distances by $O(\sqrt{n})$)**.** *Fix a similarity $S(x) = sRx + z$, with $s > 0$ and R orthogonal, and a bounded domain U. There is a field $\widehat{h}_{n,S}$ with the law of $x \mapsto h_n(Sx)$, coupled with a copy of h_n, such that, for every fixed $\varepsilon > 0$,*

$$\mathbb{P}\left[\sup_{x\in U} |\widehat{h}_{n,S}(x) - h_n(x)| \le C_{S,U,\varepsilon}\sqrt{n}\right] \ge 1 - \varepsilon, \qquad n \ge 1. \tag{31}$$

Set $X_n = D_n(A, B; U)$ and $\widehat{X}_{n,S} = sD_{\widehat{h}_{n,S}}(A, B; U)$. The latter has the law of $D_n(SA, SB; SU)$. In this coupling, on the event where the supremum is at most v,

$$s e^{-\psi v} X_n \le \widehat{X}_{n,S} \le s e^{\psi v} X_n. \tag{32}$$

The constants can be chosen uniformly over any fixed finite family of similarities and bounded domains.

Proof. Put $c = \log_2 s$. Translation and rotational invariance of white noise, radial symmetry of K, and the change of variables $(y, t) = (sRy' + z, st')$ give

$$\{h_n(Sx) : x \in U\} \overset{\text{law}}{=} \{H_{c,c+n}(x) : x \in U\}.$$

Couple $H_{c,c+n}$ and $H_{0,n}$ using one white noise defined for all positive time parameters. For $n > |c|$ their difference is

$$H_{c,c+n} - H_{0,n} = \begin{cases} H_{n,n+c} - H_{0,c}, & c \geq 0, \\ H_{c,0} - H_{n+c,n}, & c < 0. \end{cases}$$

The coarse term is a fixed-band continuous Gaussian field with a finite expected supremum on U and a Gaussian supremum tail. The fine term, after a change of scale, is a fixed-band stationary Gaussian field on a domain of diameter $O(2^n)$. Cover this domain by $O(2^{dn})$ unit cubes. On each unit cube its absolute supremum has a bounded mean and a Gaussian upper tail, as follows from (7) and the entropy calculation in Lemma 2.1. A union bound therefore bounds the largest such supremum by $C_{S,U,\varepsilon}\sqrt{n}$ with probability at least $1 - \varepsilon$. The finitely many smaller n are absorbed by increasing the constant. This proves (31). Lengths scale by s, so direct comparison of each path's density gives (32). □

Proposition 6.2 (Shape comparison)**.** *For disjoint compact sets K_1, K_2 with nonempty interiors, there is a constant $C_{K_1,K_2} < \infty$ such that*

$$b_n e^{-C_{K_1,K_2}\sqrt{n}} \leq \operatorname{Med} D_n(K_1, K_2) \leq b_n e^{C_{K_1,K_2}\sqrt{n}}, \qquad n \geq 1. \tag{33}$$

Proof. Lower bound. Let $\rho = \operatorname{dist}(K_1, K_2) > 0$ and choose $r > 0$ with $2\sqrt{d}\, r < \rho$. Cover K_1 by finitely many cubes $C_r(z_i)$ whose centers z_i belong to K_1. Since K_2 lies outside every $C_{2r}(z_i)$, every connecting path contains a crossing from $C_r(z_i)$ to $\partial C_{2r}(z_i)$ for at least one index i. Hence, pointwise,

$$D_n(K_1, K_2) \geq \min_i D_n(C_r(z_i), \partial C_{2r}(z_i); C_{2r}(z_i)). \tag{34}$$

Each distance on the right is a fixed translated and scaled reference crossing. Lemmas 6.1 and 2.2 imply that, for any prescribed fixed $\varepsilon > 0$, each is at least $b_n e^{-C_\varepsilon\sqrt{n}}$ with probability at least $1 - \varepsilon$. The fixed factor r is absorbed into the constant. Choose ε smaller than $1/(4J)$, where J is the number of covering cubes. A union bound in (34) gives the lower bound in (33) with probability at least $3/4$, and therefore gives the asserted median bound.

Upper bound. Choose points p_1, p_2 and radii $\rho_1, \rho_2 > 0$ such that the closed Euclidean balls $\overline{B}_{\rho_i}(p_i)$ lie in K_i. Put $L_0 = |p_1 - p_2|_2 > 0$. Cover ∂C_1 by finitely many closed Euclidean δ-balls with centers x_i on ∂C_1, and cover ∂C_2 similarly, with centers y_j on ∂C_2. Choose

$$0 < \delta < \min\left\{\frac{1}{10}, \frac{\rho_1}{4L_0}, \frac{\rho_2}{4L_0}\right\}.$$

All paired center separations are between 1 and $3\sqrt{d}$. For each pair (i, j), choose a similarity S_{ij} carrying x_i to p_1 and y_j to p_2. Its scale factor $s_{ij} = L_0/|x_i - y_j|_2$ is at most L_0. Thus

$$S_{ij}(\overline{B}_\delta(x_i)) \subset K_1, \qquad S_{ij}(\overline{B}_\delta(y_j)) \subset K_2. \tag{35}$$

Let $U = (-3,3)^d$ and set

$$X_n^{ij} = D_n(\overline{B}_\delta(x_i), \overline{B}_\delta(y_j); U).$$

Every reference crossing, after truncating to start on ∂C_1, has endpoints in one of these paired covers. Consequently

$$\min_{i,j} X_n^{ij} \le T_n. \tag{36}$$

This follows at the level of infima and requires no measurable choice of a minimizing path. Let M be the finite number of pairs. Since $\mathbb{P}[T_n \le b_n] \ge 1/2$, there is a deterministic pair (i_n, j_n), possibly depending on n, such that

$$\mathbb{P}[X_n^{i_n j_n} \le b_n] \ge \frac{1}{2M}. \tag{37}$$

Apply Lemma 6.1 to this pair with failure probability at most $1/(4M)$. All constants are uniform because the family of similarities is finite. By (35), the unrestricted distance between K_1 and K_2 is at most the internal distance between these mapped balls in $S_{i_n j_n} U$. Combining (32) and (37) therefore gives

$$\mathbb{P}\left[D_n(K_1, K_2) \le b_n e^{C\sqrt{n}}\right] \ge \frac{1}{4M}.$$

Here the deterministic factors s_{ij} have again been absorbed. Finally (12), applied to the unrestricted target distance, upgrades this fixed positive probability to the upper median bound in (33). □

7 Conclusion of the proof

Define

$$\boxed{\widetilde{Q}(\psi) = \frac{1}{\psi}\left(1 + \frac{\lambda}{\log 2}\right), \qquad \lambda = \lim_{n\to\infty} \frac{1}{n} \log \operatorname{Med} T_n.} \tag{38}$$

The limit exists and is finite by (30). Proposition 6.2 gives

$$\log \operatorname{Med} D_n(K_1, K_2) = \lambda n + o(n),$$

which is exactly (2). The quantile assertion follows from Lemma 2.2. For each fixed pair and each $\varepsilon > 0$, the same lemma yields

$$\sum_{n=1}^{\infty} \mathbb{P}\left[|\log D_n(K_1, K_2) - \log \operatorname{Med} D_n(K_1, K_2)| > \varepsilon n\right] < \infty.$$

Borel–Cantelli, first for positive rational ε and then by monotonicity, proves (3). This completes the proof of Theorem 1.1.

Remark 7.1 (Relation to the point-to-point normalization). The proof uses only crossing quantiles, field concentration, finite-range independence, and fixed similarities. If $Q(\psi)$ denotes the existing point-to-point exponent in the same regularization, choosing fixed $x \in K_1$ and $y \in K_2$ gives $D_n(K_1, K_2) \le D_n(x, y)$ and hence $\widetilde{Q}(\psi) \le Q(\psi)$. No reverse comparison enters the argument. In particular, the proof applies for every $\psi > 0$, including parameters for which the resulting $\widetilde{Q}(\psi)$ is negative.

Remark 7.2 (The geometric scope). The upper comparison in Proposition 6.2 allows the connecting path to use all of $\mathbb{R}^d$, exactly as in equation (1.9) of [1]. The local object used to define the exponent is an annular crossing, for which a path can be clipped at the first outer-boundary hit. The argument does not need an upper concatenation estimate for paths constrained to cross an arbitrary prescribed box.

References

[1] J. Ding, E. Gwynne, and Z. Zhuang. *Percolation of thick points of the log-correlated Gaussian field in high dimensions*. arXiv:2410.22075, version 2, 2026. https://arxiv.org/abs/2410.22075. The field and metric are defined in (1.5) and (1.7); the set-to-set exponent is defined in (1.9).

[2] J. Ding, E. Gwynne, and Z. Zhuang. *Tightness of exponential metrics for log-correlated Gaussian fields in arbitrary dimension*. arXiv:2310.03996, version 5, 2026. https://arxiv.org/abs/2310.03996. Lemma 2.7 gives logarithmic Gaussian concentration in the smooth-kernel setting; Section 3 establishes the point-to-point exponent.